

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

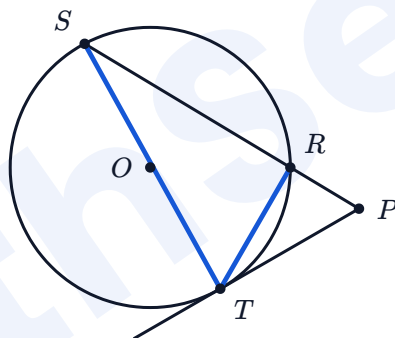
Chapter 18: Tangents and Intersecting Chords

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Tangent and secant from an external point

1. PT is a tangent to a circle at T , and a line through P cuts the circle at R and S , with R nearer to P . (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT . **ICSE 2025, 3 marks**



SOLUTION

(a) In $\triangle PTR$ and $\triangle PST$: $\angle P$ is common, and $\angle PTR = \angle PST$ (angle between tangent and chord equals the angle in the alternate segment). So $\triangle PTR \sim \triangle PST$ (AA).

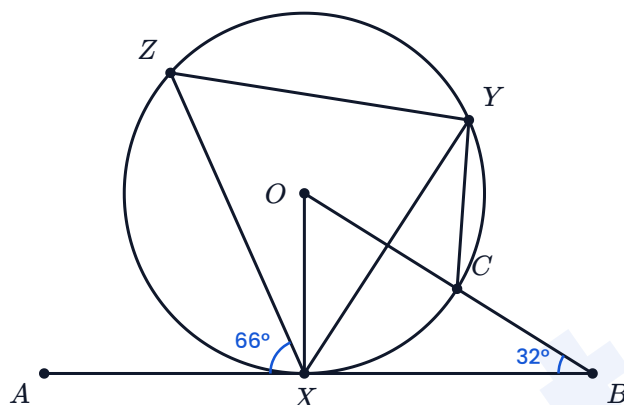
(b) Corresponding sides are proportional: $\frac{PT}{PS} = \frac{PR}{PT}$, so $PT^2 = PR \times PS$.

(c) $PT^2 = 4 \times 16 = 64$

Answer: $PT = 8$ cm

2 The alternate segment theorem

2. X, Y, Z and C lie on a circle with centre O . AB is the tangent at X , OB cuts the circle at C , and $ZY = XY$. Given $\angle OBX = 32^\circ$ and $\angle AXZ = 66^\circ$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$.
ICSE 2025, 4 marks



SOLUTION

- (a) $OX \perp AB$ (radius and tangent). In $\triangle OXB$: $\angle BOX = 90^\circ - 32^\circ = 58^\circ$.
 (b) The arc CX subtends $\angle COX = 58^\circ$ at the centre and $\angle CYX$ on the circle: $\angle CYX = \frac{58^\circ}{2} = 29^\circ$.
 (c) By the alternate segment theorem, $\angle ZYX = \angle AXZ = 66^\circ$.
 (d) $ZY = XY$, so $\angle YXZ = \frac{180^\circ - 66^\circ}{2} = 57^\circ$. Also $\angle OXZ = 90^\circ - 66^\circ = 24^\circ$. So $\angle OXY = 57^\circ - 24^\circ = 33^\circ$.

Answer: (a) 58° (b) 29° (c) 66° (d) 33°

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